SHIZHONG TAN

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RESEARCH PROFILE

My research centers on solving the bottleneck of **high-precision robotic milling for large-scale aerospace components**. Working at the intersection of robot dynamics, machine learning, and control engineering, I have developed a comprehensive framework to enhance robotic contour accuracy under extreme dynamic conditions.

Core Academic Competencies:

- **Robot Dynamics and Process Optimization:** Established a Lie-theory-based unified kinematics-dynamics modeling framework considering joint compliance to mathematically decode complex robotic dynamics. Building upon this, formulated a fast-calculation method for tool-tip force-induced deformation and cutting forces, enabling the simultaneous optimization of robot redundancy angles and tool orientations to minimize tool tip force-induced deformation error.
- **Intelligent Error Compensation and Vision Fusion:** Proposed multi-feature hybrid deep learning frameworks to predict and compensate for pose-dependent trajectory and contour errors. By fusing the robot's inherent kinematic features with in-situ structured-light visual measurements, developed a neural-network-driven method that directly predicts machining deviations. This effectively bypasses the reliance on offline Coordinate Measuring Machine inspections, enabling closed-loop precision control.
- **System Integration and Hardware Innovation:** Led the electromechanical design and decoupling control of a novel two-DOF active vibration suppression physical system. By tackling strong pose-dependent dynamic coupling and time-varying modal characteristics, the hardware-algorithm co-design significantly mitigates milling vibration, bridging the gap between theoretical dynamics and industrial aerospace manufacturing applications.

Moving forward, I aim to leverage my expertise in robot dynamics and AI-driven control to explore broader applications in advanced robotic manipulation, compliant control, and next-generation smart manufacturing systems.

EDUCATION

Huazhong University of Science and Technology, Wuhan, HuBei, China — Sept. 2020 - Expected June 2026
Doctor of Mechatronic Engineering
Dissertation: Research on Methods for Improving Contour Accuracy in Robotic Milling of Complex Deep-Cavity Casing
Supervisors: Prof. Jixiang Yang and Prof. Han Ding

Central South University, ChangSha, HuNan, China — Sept. 2016 - June 2020
Bachelor of Mechanical Design and Manufacturing Engineering

PUBLICATIONS

Authored **5** peer-reviewed journal articles, including **4 as the first author** in top-tier robotics and manufacturing journals such as **IJRR** and **RCIM**.

Published papers

1. **Shizhong Tan**, Xu Tang, Jixiang Yang and Han Ding. "Design of a novel two-DOF robotic active vibration suppression system based on dynamic decoupling and pose-dependent dynamics." ***The International Journal of Robotics Research*** (2025): 02783649261423559. (IF: 5.0, Top Tier in Robotics)
2. **Shizhong Tan**, Jixiang Yang, and Han Ding. "A prediction and compensation method of robot tracking error considering pose-dependent load decomposition." *Robotics and Computer-Integrated Manufacturing* 80 (2023): 102476. (IF: 11.4, Cited: 81)

3. **Shizhong Tan**, Jixiang Yang, and Han Ding. "Processing accuracy improvement of robotic ball-end milling by simultaneously optimizing tool orientation and robotic redundancy." *Robotics and Computer-Integrated Manufacturing* 93 (2025): 102904. (IF: 11.4)
4. **Shizhong Tan**, Chengxing Wu, Jixiang Yang, and Han Ding. "A contour error prediction method for tool path correction using a multi-feature hybrid model in robotic milling systems." *Robotics and Computer-Integrated Manufacturing* 93 (2025): 102936. (IF: 11.4)
5. Yinxin Guan, Jixiang Yang, **Shizhong Tan**, and Han Ding. "A data-driven rolling optimization method for trajectory tracking error prediction of CNC machine tools." *Science China Technological Sciences* 68.1 (2025): 1120301. (IF: 4.9)

Manuscripts Under Review

1. **Shizhong Tan**, Jixiang Yang and Han Ding, "Lie-Theory-Based Robots Unified Kinematics–Dynamics Modeling and Iterative Parameter Identification Considering Joint Compliance." *IEEE Transactions on Robotics*.
2. Hongwei Sun, **Shizhong Tan**, Jixiang Yang and Han Ding, "A novel data-driven prediction and compensation method for robot contour errors considering the joint dynamics." *IEEE/ASME Transactions on Mechatronics*.

CONFERENCES

Oral Presentation

1. **Shizhong Tan**, Jixiang Yang, and Han Ding. (2023) Positioning accuracy improvement of robotic ball-end milling by simultaneously optimizing the tool orientation and robotic redundance." The 20th International Manufacturing Conference in China. Chongqing, China. (Second Prize Paper).
2. **Shizhong Tan**. (2024). Unified optimization of tool orientation and redundancy angles in robotic ball-end milling. Doctoral Workshop 2024 (Session III), China Graphics Society. Online.

Poster Presentation

1. **Shizhong Tan**, Jixiang Yang, and Han Ding. (2025). Active vibration suppression system design for industrial robots based on dynamic decoupling and pose-dependent dynamics. In The 6th Annual Conference of China Robotics Society. Changsha, China.

PATENTS

Co-inventor (Student First Inventor) of **5** Chinese National Invention Patents. These patents focus on robotic precision manufacturing, adaptive vibration suppression, and digital twin systems.

1. Jixiang Yang, **Shizhong Tan**, and Han Ding. Offline joint error compensation method, system, and terminal for gravity pose decomposition of industrial robots. Chinese Invention Patent No. CN114131611B, Granted.
2. Jixiang Yang, **Shizhong Tan**, and Han Ding. Optimization method of tool orientation and redundancy angles in robotic ball-end milling. Chinese Invention Patent No. CN117908466B, Granted.
3. Jixiang Yang, **Shizhong Tan**, and Han Ding. Active vibration suppression method and system for robotic milling considering pose-dependent modal dynamics. Chinese Invention Patent No. CN118700153B, Granted.
4. Jixiang Yang, **Shizhong Tan**, and Han Ding. Decoupling control method and system for a two-degree-of-freedom robotic vibration suppressor. Chinese Invention Patent, Application No. CN119610116B, Granted.
5. Chengxing Wu, **Shizhong Tan**, Jixiang Yang and Han Ding. Robotic milling path error prediction method, system, and storage medium. Chinese Invention Patent, Application No. CN119691701B, Granted.

RESEARCH FUNDING EXPERIENCE

1. The National Key Research and Development Program of China. Rigid–Flexible Coupled Dynamic Modeling and High-Precision Identification of Industrial Robots, Grant No.2023YFB4703800.

Role: Formulated a Lie-theory-based unified kinematics-dynamics modeling framework considering joint compliance. Led the design of a novel 2-DOF active vibration suppressor for robotic milling. By implementing control force decoupling and an adaptive control gain strategy based on pose-dependent modal parameters, achieved robust and stable vibration suppression across the robot's entire workspace.

2. The Basic Science Center Program of the National Natural Science Foundation of China. Robotic Intelligent Manufacturing, Grant No. 52188102.

   Role: Identified and extracted the pose-dependent payload characteristics of the robotic system. Engineered a neural-network-driven prediction model to dynamically forecast joint servo errors. The model significantly enhanced the robot's overall servo tracking precision by implementing proactive error compensation.

3. The Joint Funds of the National Natural Science Foundation of China. Accuracy and Surface Quality Control in Robotic Compliant Machining of Large Complex Surfaces, Grant No. U24A20276.

   Role: Defined and quantified the tool-tip force-induced deformation error. By developing a fast-calculation method for cutting forces under varying tool postures, the tool orientation and robotic redundancy angles were synergistically optimized. This approach effectively minimized deformation errors induced by spindle gravity and cutting forces during heavy-duty milling.

4. The Young Scientists Fund of the National Natural Science Foundation of China. In-situ Detection and Compensation of Contour Error in Robotic Milling of Large Deep-Cavity Components, Grant No. 52305535.

   Role: Proposed a multi-modal data fusion framework integrating the robot's inherent physical machining features with in-situ structured-light visual measurements. This enabled the closed-loop compensation of contour errors, substantially elevating the final machining accuracy without relying on offline coordinate measurements.

SKILLS

- **Programming Languages:** Python, C++, MATLAB, PyTorch
- **Software:** ROS, UG, SolidWorks, Linux
- **Hardware Platform:** KUKA/ABB/UR industrial robots, Force/Acceleration sensors, Laser trackers

REFERENCES

1. **Prof. Han Ding**
   Professor, School of Mechanical Science and Engineering, Huazhong University of Science and Technology
   Email: dinghan@hust.edu.cn
   Relationship: Ph.D. Co-supervisor

2. **Prof. Jixiang Yang**
   Professor, School of Mechanical Science and Engineering, Huazhong University of Science and Technology
   Email: jixiangyang@hust.edu.cn
   Relationship: Ph.D. Supervisor